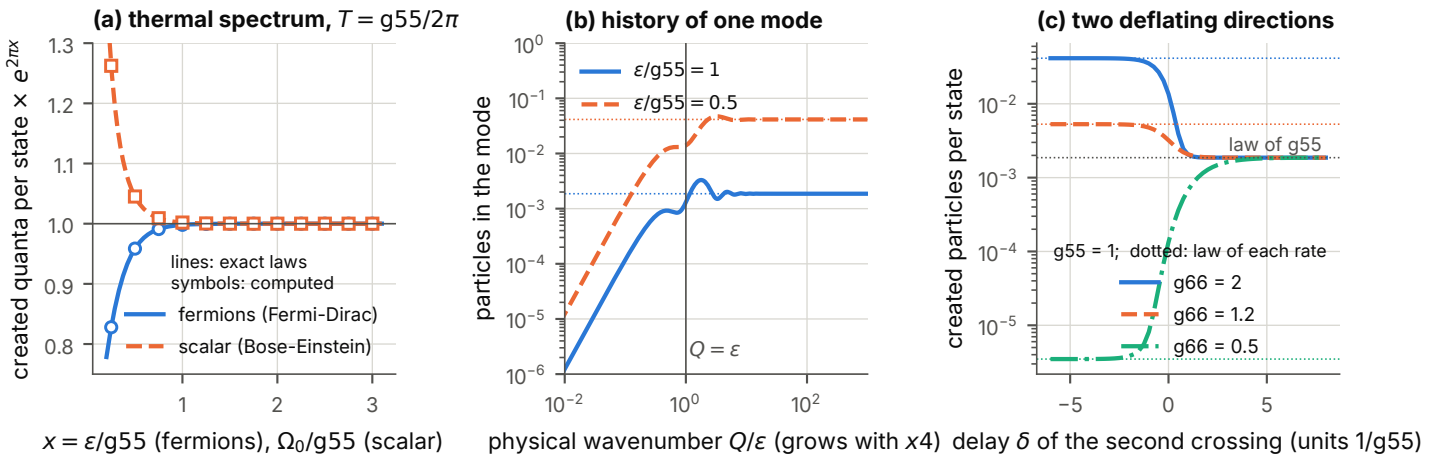
(a) thermal spectrum, $T = g55/2\pi$
created quanta per state × $e^{2\pi x}$
1.3
1.2
1.1
1.0
0.9
0.8
1
2
3
lines: exact laws
symbols: computed
fermions (Fermi-Dirac)
scalar (Bose-Einstein)
$x = \varepsilon/g55$ (fermions), $\Omega_0/g55$ (scalar)
(b) history of one mode
particles in the mode
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
$\varepsilon/g55 = 1$
$\varepsilon/g55 = 0.5$
$Q = \varepsilon$
10^{-2}
10^{0}
10^{2}
physical wavenumber Q/ε (grows with $x4$)
(c) two deflating directions
created particles per state
10^{-2}
10^{-3}
10^{-4}
10^{-5}
law of g55
g55 = 1; dotted: law of each rate
g66 = 2
g66 = 1.2
g66 = 0.5
−5
0
5
delay δ of the second crossing (units 1/g55)